

Generation Interconnection Request System Impact Study Report

For Queue Project Q18-1009

Lone Tree 138 kV
January 22, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1009. The Interconnection Customer proposes a 219.4 MW solar photovoltaic facility interconnecting at the Lone Tree 138 kV substation in the Virginia area. The request was received on June 04, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 327 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$1,747,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Kerr Dam to Mecklenburg Power Sta 345 kV line	Bear to Mecklenburg Power Station 345 kV	113.9	1,198	8.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Kerr Dam to Mecklenburg Power Station 345 kV	Rebuild line with higher-capacity conductor	\$31,150,000	6% (\$1,747,000)
	Total allocated to project		\$1,747,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.